

① DETERMINE THE VALUE OF x , GIVEN $\log_x \left(\frac{1}{243} \right) = 5$
WHERE x IS THE 'BASE'

② SOLVE y IN TERMS OF x IF $\log_7 y = 2 \log_7 5 + \log_7 x + 3$

BASE = 7

③ SOLVE y IN TERMS OF x IF $10^{\frac{y}{x+1}} = 3$

④ PLOT $xy^4 = 16$ USING LOG-LOG PAPER

⑤ PLOT $y = 5x^3$ ON SEMILOG PAPER

⑥ EVALUATE $5 \left(10^{2 \log 3} \right)$

⑦ THE CURRENT i (in μA) IN A MICROCHIP CIRCUIT IS

$i = 2.5 \left(2^{-2500t} \right)$ WHERE t IS TIME IN NS (nano sec)
AFTER THE CURRENT IS TURNED OFF. HOW LONG
DOES IT TAKE THE CURRENT TO BE $0.25 \mu A$?

- ⑧ THE AMOUNT OF ALCOHOL IN A PERSON'S BLOODSTREAM IS GIVEN BY $A = A_0 e^{-kt}$,

WHERE A_0 IS AN INITIAL AMOUNT (mg alcohol/mL BLOOD),

t IS THE TIME (IN HRS) AFTER DRINKING THE ALCOHOL,

AND k IS A CONSTANT DEPENDING ON THE PERSON.

IF $A_0 = 0.24$ mg/mL, AND $A = 0.16$ mg/mL IN 2.0 HRS, HOW LONG SHOULD THE PERSON WAIT TO DRIVE IF THE LEGAL LIMIT IS 0.08 mg/mL?

- ⑨ A STATE LOTTERY PAYS \$500 FOR A \$1 TICKET IF A PERSON PICKS THE CORRECT THREE-DIGIT NUMBER DETERMINED BY THE RANDOM DRAW OF THREE NUMBERED BALLS. THE PROBABILITY OF A 50% CHANCE OF WINNING IN x DRAWING IS $(1 - 0.999^x) = 0.5$. FOR HOW MANY DRAWINGS DOES A PERSON HAVE TO BUY A TICKET TO HAVE A 50% CHANCE OF WINNING?

- ⑩ THE TIME t (IN SEC) TO CHEMICALLY CHANGE 5 KG OF A CERTAIN SUBSTANCE INTO ANOTHER IS GIVEN BY $t = -5 \log\left(\frac{5-x}{5}\right)$, WHERE x IS THE NUMBER OF KILOGRAM THAT HAS BEEN CHANGED AT ANY TIME. SKETCH THE GRAPH OF t VS x .

- ⑪ TO FIND NUMBER n OF YEARS THAT AN INITIAL VALUE A OF EQUIPMENT TAKES TO HAVE SALVAGE VALUE S , THE EQUATION $n = \frac{\log S - \log A}{\log(1-d)}$ IS USED, d = ANNUAL RATE OF DEPRECIATION. SOLVE FOR d .

- (12) THE EFFICIENCY η OF A GASOLINE ENGINE AS A FUNCTION OF ITS COMPRESSION RATIO r IS GIVEN BY:

$$\eta = 1 - r^{(1-\gamma)}, \text{ WHERE } \gamma \text{ IS A CONSTANT.}$$

FIND γ FOR $\eta = 0.55$ AND $r = 7.5$